TensorFlow Environment Setup

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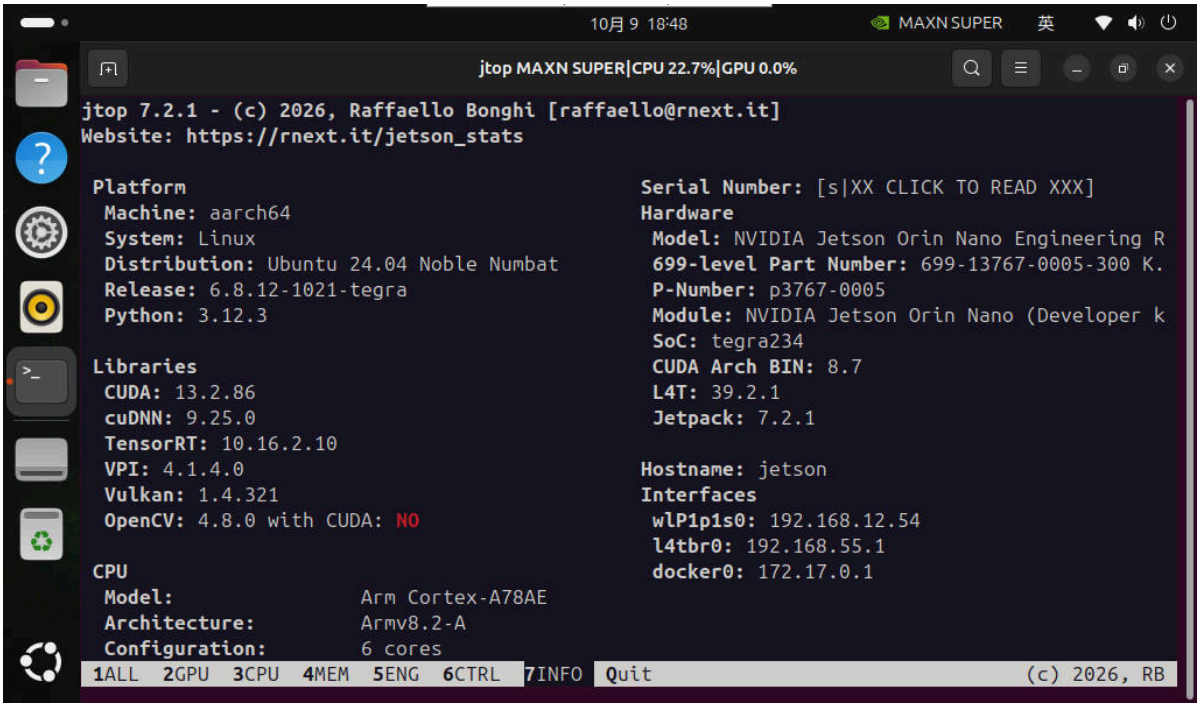
References

> **Note on pip installation:** Ubuntu 24.04 marks the system Python as "externally managed" according to PEP 668, so installing packages directly with pip may report `externally-managed-environment`. The `--break-system-packages` option allows pip to install packages in that environment; virtual environments usually do not need this option, but it is kept here to keep the commands in this tutorial consistent.

> The factory system already has the relevant environment set up, so the installation steps can be skipped.

TensorFlow is an end-to-end machine learning and deep learning framework developed and open-sourced by Google, used to build and train various types of machine learning models.

1. System Information



> **Check the CUDA version**

```
nvcc --version
```

2. Installing TensorFlow

The current system uses TensorFlow `2.16.1`. For installation, choose either the offline or the online method; this version has only been verified on CPU in the current Jetson environment and does not provide GPU acceleration.

2.1. Offline Installation

On a device with internet access, download the TensorFlow 2.16.1 wheel for Python 3.12, Linux ARM64, along with its dependencies, then copy the files to the Jetson.

Download URL: https://pypi.org/project/tensorflow/2.16.1/#files

```
cd Downloads/
```

```
python3 -m pip install --user --break-system-packages --no-index --find-links . tensorflow==2.16.1
```

2.2. Online Installation

```
python3 -m pip install --user --break-system-packages tensorflow==2.16.1
```

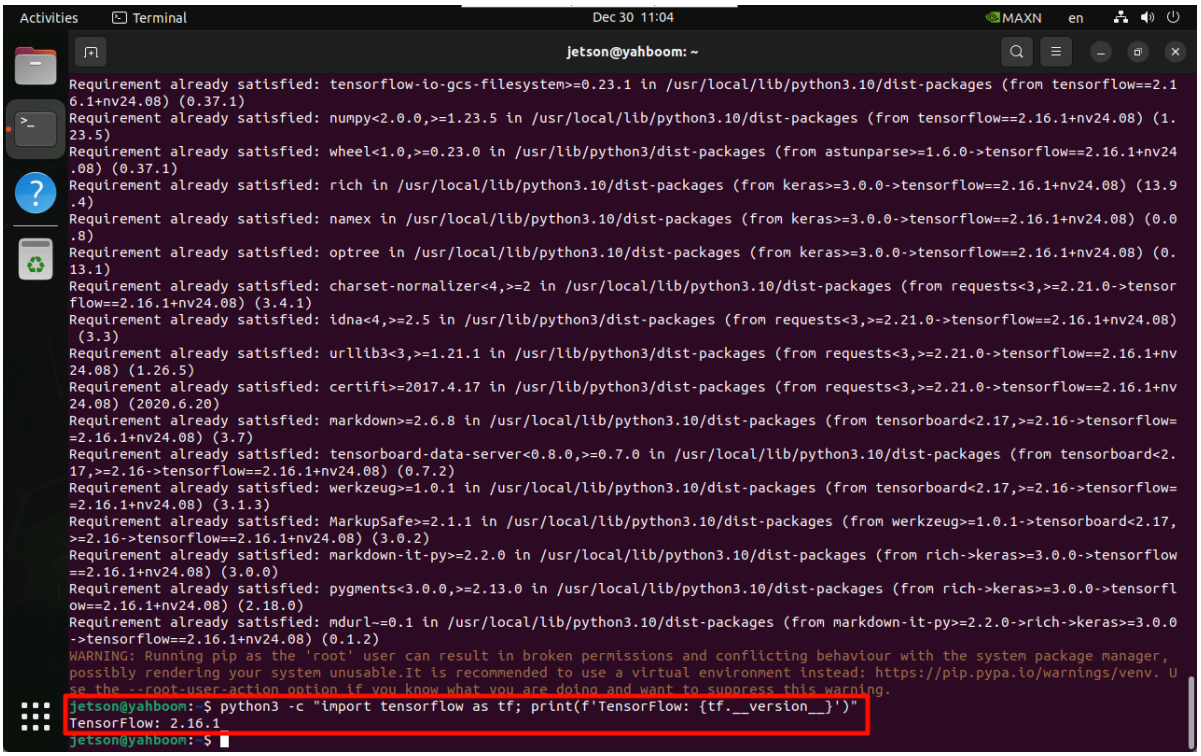
2.3. Installing Numpy

The current system's NumPy `1.26.4` is provided by APT; do not downgrade it to `1.23.5` with pip.

```
sudo apt install python3-numpy
```

3. Verifying the Installation

```
python3 -c "import tensorflow as tf; print(f'TensorFlow: {tf.__version__}')"
```



References

https://docs.nvidia.com/deeplearning/frameworks/install-tf-jetson-platform/index.html

https://pypi.org/project/tensorflow/2.16.1/